Bogleheads

Risk and return: an introduction

Risk and return is a complex topic. There are many types of risk, and many ways to evaluate and measure risk. In investing, one widely used definition of risk is:[1][note 1]

> ... the uncertainty that an investment will earn its expected rate of return.
>
> — Frank K. Reilly, *Investment Analysis and Portfolio Management*

Notice that this definition does not distinguish between loss and gain. Investors typically think of risk as the possibility that their investments could lose money.[2] They are likely to be quite happy with an investment return that is greater than expected - a "positive surprise." However, because risky assets generate negative surprises as well as positive ones, it is reasonable to define risk as the **uncertainty** of returns. Greater uncertainty means that it is more likely that the investment will generate larger gains, as well as more likely that it will generate larger losses (in the short term), and higher or lower accumulated value (in the long term.)

In financial planning, you need to consider your investment goal to define which risk matters to you. For example, if you want an acceptable retirement income, you need to build an investment portfolio that generates an *expected* return sufficient to create that level of income.

But because it is **uncertain** that your portfolio will earn its *expected* long-term return, its long-term *realized* return may be less than your expected return. In that case, your available retirement funds will be less than you need - that is, *you might outlive your investment portfolio*. This is an example of "shortfall risk." The **magnitude** and **consequences** of this deserve special consideration.[3]

However, because it is **uncertain**, your portfolio could *also* have a *realized* return that is *higher* than your expected return; the *investment portfolio might "outlive" you*.

As a result, shortfall risk is generally just one part of considering risk as the *uncertainty of investment return*.

Risk as the uncertainty of returns

Historical distributions of returns in three major asset classes — cash, bonds, and stocks — demonstrate the uncertainty in investing.[4]

The term *cash* usually refers to money market securities and money in bank accounts. Vanguard refers to these types of assets as *short-term reserves*. There is very high certainty in the return that you would earn on an investment in money market securities such as Treasury bills (T-Bills) or short-term certificates of deposit (CDs). Similarly, there is fairly high certainty in the return that you would earn over a short period in a money market fund. Money market fund holdings consist of T-Bills, CDs, and other money market securities.[5]

For an individual investor, a federally-insured bank account also provides a high degree of certainty in the short-term return.

Even over longer time periods, the returns earned by money market securities fall into a relatively narrow range.

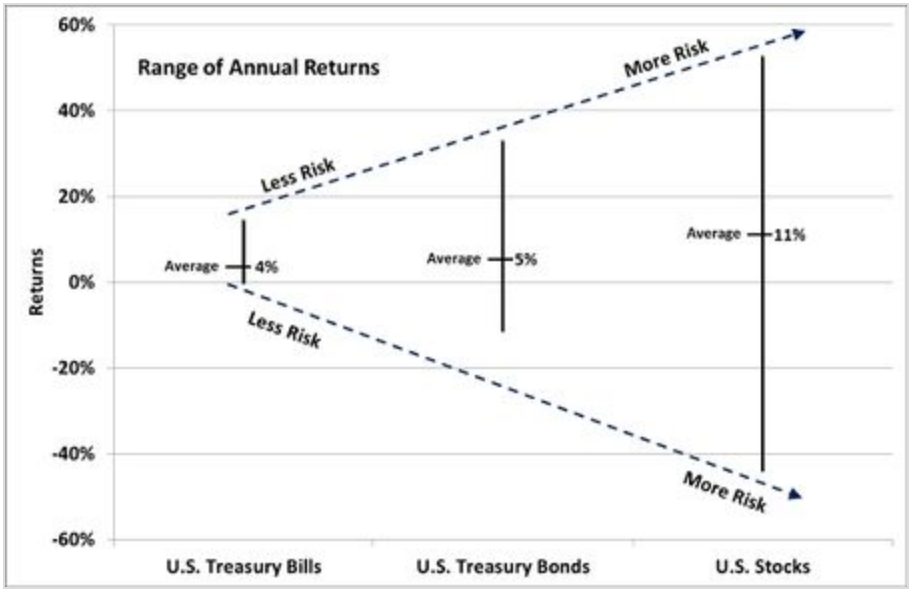


Figure 1. Risk vs. Return: 1928 - 2011[6]

The annual returns on 3-month T-Bills (representing "cash") were in the range of 0% to 15% during the 84 year period from 1928 through 2011. In Figure 1, the vertical bar on the far left (above the label "U.S. Treasury Bills) represents this range of returns. Also note the tick mark showing the average annual return of 4%.

The middle vertical bar in Figure 1 represents the range of annual returns on 10-Year U.S. Treasury Bonds over the same time period. There is a larger range (dispersion) of returns, from about -11% to +33%. Also note the somewhat larger average annual return of 5%.

The far right vertical bar in Figure 1 represents the range of annual returns on U.S. stocks (as represented by stocks in the S&P 500) over the same time period. There is a much larger dispersion of returns, from about -44% to +53%. Also note that the average annual return of 11% is significantly higher.

As the chart indicates, the return on bonds is less certain than the return on cash, and the return on stocks is less certain than the return on bonds. As a result, bonds are considered riskier than cash, and stocks are considered riskier than bonds.[note 2][7][8][9][10][11]

The chart also shows that historically, over the relatively long time period of 84 years from 1928 through 2011, higher risk has been rewarded with higher **average** annual returns, at least in the United States. This pattern is not unique to the U.S., but it is not the same in all countries.[12]

Relationship between risk and return

Investors are risk averse; that is, given the same expected return, they will choose the investment for which that return is more certain. Therefore, they demand a higher expected return for riskier assets. A higher *expected* return does not guarantee a higher *realized* return. Returns on risky assets are uncertain, so an investment may not earn its expected return.

Although the charts in Figure 1 show historical (realized) returns rather than expected (future) returns, they show the relationship between risk and return. The mean (average) annual return increases as the dispersion of returns increases.

This demonstrates one of the most fundamental axioms of investing:

> *Risk and return are inextricably related. You can generally achieve higher returns only by taking more risk, but because the risk exists, the higher expected returns may not result in higher realized returns.*

If inflation is considered, even money market securities have some risk. They may not achieve the expected real (inflation-adjusted) return. Unexpected inflation may reduce the *real* return below the *expected* return of the money market investment. You can eliminate uncertainty in real returns by investing in inflation-indexed securities, such as Treasury Inflation Protected Securities (TIPS) and Series I Savings Bonds (I Bonds). In return for this reduction of uncertainty, you need to accept lower expected returns. Even inflation-linked securities have risks; for example, TIPS have interest-rate risk, re-investment risk, and liquidity risk. No investment is truly risk-free.

Managing portfolio risk

Diversification

You can reduce risk through diversification.

The risk of investing in a single risky security, such as a stock or corporate bond, is very high due to the company-specific risks. Any number of unfortunate events could impact the rate of return. In the worst possible case, the company could go bankrupt, and you could lose the entire value of the investment. Company-specific risk is generally referred to as unsystematic risk or non-systematic risk. Other names are unique-risk, firm-specific risk, or diversifiable risk.

You can eliminate unsystematic risk by holding a broad portfolio of risky assets; that is, many different securities in many different industries. This is easily done by owning a total market stock or bond index fund. Unsystematic risk is risk that can be "diversified away."

The risk that remains after diversifying away unsystematic risk is systematic risk. Other names are market risk or non-diversifiable risk. A total stock or bond market fund has systematic risk. This is risk impacting an entire asset class, such as when rising real interest rates impact the entire bond market.

In an efficient market, assets with known systematic risks will be priced lower, and so compensate with higher expected returns. This expected relationship only applies to systematic risks. There is no reward for incurring unsystematic risk, and you may therefore use broad diversification without reducing the expected return of your portfolio.

Asset allocation

Main article: Asset allocation

After diversification, the next step in managing your portfolio risk is asset allocation.

In **theory**, asset allocation means selecting an appropriate mix of risk-free assets and risky assets. Optimally, the risky portion of your portfolio includes all risky assets; for example, stocks, bonds, real estate, and so on. A 30-day T-Bill is most commonly used to represent the risk-free asset.

In **practice**, short-term, high quality bonds are considered to be relatively risk free, and so are typically considered part of the (relatively) risk-free portion of your portfolio. Although a bond fund that includes riskier bonds (for example, corporate bonds and bonds with longer maturities) is technically a risky asset, the risk is low in comparison to stocks. When investing according to the Bogleheads® investment philosophy, you would generally use broad-based, short- to intermediate-term bond funds for the low-risk portion of your portfolio, and broad-based stock funds for the risky portion of your portfolio.

Selecting the appropriate asset allocation (ratio of risky assets to low-risk assets) *is essential* to designing a portfolio that matches your ability, willingness, and need to take risk.[13]

An illustrative example

Main article: Risk and return: application

For an illustrative example which shows how you can use these ideas to plan a portfolio, see Risk and return: application.[note 3]

Specific types of risk

Below are descriptions of different types of investment risks.[14] A mutual fund prospectus will often outline these types of risk.

Portfolio theory makes an important distinction between two types of risks:

- **Unsystematic risk**: The measure of risk associated with a particular security; also known as **diversifiable risk**. You can reduce unsystematic risk (http://www.investopedia.com/terms/u/unsystematicrisk.asp) by holding a diversified portfolio of many different stocks in many different industries.
- **Systematic risk**: Also known as **market risk**. Because of market volatility, all investors face systematic risk (https://www.investopedia.com/terms/s/systematicrisk.asp). This risk cannot be diversified away. This is the type of risk most people are referring to when they casually use the term "risk" when discussing investments.

Some additional risks faced by all investments include:

- **Liquidity risk**: The risk that an asset cannot be sold when needed, or in sufficient quantity, because opportunities are limited. Treasury securities (except for inflation protected Treasury bonds) have the least liquidity risk.
- **Political risk**: The risk to an investment from changes in the law or political regime. Potential changes in tax law or changes in a country's structure of governance are sources of political risk.
- **Inflation risk**: Stocks, bonds and cash are all subject to the risk that investments will not keep pace with inflation. You can reduce this risk by investing in inflation-protected Treasury bonds, such as TIPS or I-bonds.

- **Financial risk**: The risk from the capital structure of a firm. Corporate debt magnifies financial risk to a company's stocks and bonds.
- **Management risk**: Investors using actively managed funds have the risk that fund or portfolio managers will under-perform benchmarks because of their management decisions or style. You can avoid this risk by selecting passively-managed index funds.

Bond investors also face the following major risks:

- **Interest rate risk**: The risk associated with changes in asset price due to changes in interest rates. Bonds and bond funds face this type of risk. As interest rates rise, prices on existing bonds decline and vice versa. Interest rate risk is greater for bonds with longer maturities, and lower for bonds having short maturities.
- **Credit risk**: The risk of default. Holders of corporate and municipal bonds face this risk.

Other risks applicable to bond investments include:

- **Call risk**: The risk that after a decline in interest rates, a bond issuer could redeem a bond early, forcing you to find a replacement investment that may not pay as well as the original bond.
- **Reinvestment risk**: The risk that earnings from current investments will not be reinvested at the same rate of return as current investment yields. Coupon payments from a bond may suffer reinvestment risk if you cannot reinvest them at the same rate as the bond's yield.
- **Currency risk**: Investors in international stocks and bonds are also exposed to the risk from changes in currency exchange rates. Investments in currencies other than the one in which you purchase most goods and services are subject to currency risk.

Individual investors are exposed to two additional risks:

- **Longevity risk**: The risk you will outlive your money.
- **Shortfall risk**: The risk the portfolio will not provide sufficient returns to meet your goal(s).[15] John Bogle states:[16]

> Consideration of bonds as an important asset class implicitly requires us to recognize, as I quoted in "Common Sense on Mutual Funds", that “risk is not short-term volatility, for the long term investor can afford to ignore that. Rather, because there is not a predestined rate of return, only an expected one that may not be realized, the risk is the possibility that, in the long-run, stock returns will be terrible." Put another way, the risk is that the investment portfolio might not provide its owner—individual or institution—with adequate cash to meet future requirements for essential outlays. In short, that the investor will lose a ton of money, just when it is needed the most.
>
> — John Bogle

Notes

1. At some time in the future, the *actual* return will be one of many possible outcomes. The various outcomes have some probability of occurring. The expected return is just the average of these possible returns weighted (multiplied) by the respective probabilities of occurring.
2. Standard deviation is a statistical measure that is often used to quantify the dispersion (variation) of investment returns which provides a standardized way to compare risk for different investments. Standard deviation increases as dispersion of annual returns increases. This is illustrated by the

standard deviation of the asset classes shown in the returns history shown in Figure 1.

- US Treasury bills standard deviation 3.04%
- US 10-year Treasury bonds standard deviation 7.78%
- US S&P 500 index of stocks standard deviation 20.11%

Standard deviation of annual returns is most useful for measuring risk over shorter time periods. For measuring risk over longer time periods, the dispersion of possible **cumulative** returns is a better measure of risk. This is because over many years, a relatively small difference in annualized rate of return can result in a large difference in cumulative returns. The cumulative return on your investments at a specified future time is referred to as terminal wealth, and its possible dispersion is referred to as *terminal wealth dispersion*.

3. For those who have never experienced a significant downturn in the market, please read Bogleheads forum post: "Bear Markets" (https://www.bogleheads.org/forum/viewtopic.php?p=329947#p329947). November 15, 2008 by Taylor Larimore. Here is an excerpt:

> Figures cannot convey the horrifying and debilitating effects of a deep and long bear market. You watch in agony as month after month your life savings evaporate before your eyes. Gloom and doom articles are in the media, radio, (and now TV and internet). Nearly everyone else is selling. You have no idea when, or if, your portfolio will stop losing money.
>
> Your friends and relatives urge you to sell. Nearly all financial experts recommend "sell". You are ridiculed for trying to hold on. You begin to have self-doubt. Despair sets in. Buying stocks is unthinkable. Suicides increase.
>
> That's what a bad bear market is like.

See also

- Risk and return: application
- Risk tolerance

References

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15. "Vanguard's Investment Philosophy We Believe #7" (https://personal.vanguard.com/us/LiteratureRequest?FW_Activity=ViewOnlineActivity&vendorID=S537&FW_Event=start&search_mode=true&cbdForceDomain=false).
16. John Bogle. "Risk and Risk Control in an Era of Confidence (or is it Greed?)" (https://johncbogle.com/speeches/JCB_NE_Pension_4-00.pdf) (PDF). (Footnote 3) Lawrence Siegel, the treasurer of the Ford Foundation, writing in the Journal of Portfolio Management. Retrieved 4 April 2012.

External links

- Asset allocation: Key to your investment climate (https://investor.vanguard.com/investor-resources-education/how-to-invest/asset-allocation), from Vanguard.
- The file used to generate the Figures is available as a free download from Google Docs: Risk - Historical Performance of Bonds and Stocks.xlsx (https://docs.google.com/open?id=0B8VSNcUR-B6Idk9welFiMDhRX3UybzZxdjdseDVwdw).

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